



IAM Project - User Lifecycle Management (Banking Environment)



1. Project Overview:

This project demonstrates a simulated Identity & Access Management (IAM) User Lifecycle workflow aligned with banking security, audit, and compliance requirements. The objective is to showcase how IAM teams manage user access securely throughout the employee lifecycle while preventing unauthorized access, privilege creep, and insider threats.

This project covers Joiner, Mover, and Leaver (JML) processes, access reviews, SLA adherence, and escalation handling as expected in an IAM L1 support role.

2. Business Context:

Industry: Banking & Financial Services

Environment: Enterprise IAM operations

Security Focus:

- Least Privilege
- Audit & Compliance
- Insider Risk Reduction
- SLA-driven execution



3. Scope of the Project:

- User Provisioning (Joiner)
- Role Change Handling (Mover)
- User De-Provisioning (Leaver)
- RBAC-based Access Assignment
- Approval Validation
- Access Reviews
- Audit Documentation
- Escalation Handling

4. Tools & Systems (Simulated / Foundational):

- Microsoft Entra ID (Azure AD) – identity objects & access control
- ServiceNow – incident & service request handling
- RBAC Groups – role-based access
- Approval Workflow – manager & application owner
- IAM Runbooks / SOPs

Note: This project focuses on process accuracy and security controls, not tool depth.

5. IAM Objects Involved:

- Users
- Security Groups
- Roles
- Applications
- Service Requests
- Audit Logs



6. User Lifecycle Workflows

6.1 Joiner Process (New Employee):

Trigger: New employee onboarding request

Steps:

Service Request received via ServiceNow

Validate:

- Manager approval
- Role requested
- Map role to RBAC security groups
- Provision access via group membership
- Verify access assignment
- Document actions
- Close ticket within SLA

Security Controls:

- Least privilege applied
- No direct user-level access
- Group-based provisioning only



6.2 Mover Process (Role / Team Change):

Trigger: Internal role change

Steps:

- Receive role change request
- Validate new role approval
- Identify existing access
- Remove outdated permissions
- Assign new RBAC groups
- Confirm access accuracy
- Document changes
- Risk Addressed:
- Privilege creep
- Unauthorized access

6.3 Leaver Process (Employee Exit):

Trigger: HR termination event

Steps:

- Verify termination status
- Disable user account
- Remove group memberships
- Revoke application access
- Monitor for post-exit activity
- Escalate if access remains active
- Document completion



Critical Risk:

- Insider threat
- Audit violation

7. Access Review Process:

Frequency: Periodic (Quarterly / Bi-Annual)

Participants:

- Application Owners
- Data Owners
- Managers

Process:

- Review existing access
- Identify excess permissions
- Approve or revoke access
- IAM executes revocation
- Document review outcomes

Purpose:

- Prevent privilege creep
- Maintain audit readiness



8. Runbook & SOP Adherence:

- All actions follow documented IAM runbooks
- Deviations require escalation
- No access without documented approval
- All steps are logged for audit

9. SLA & Escalation Handling:

- SLA acknowledgment mandatory
- High-risk delays escalated to L2
- Escalation triggered when:
 - Approval missing
 - Access risk identified
 - Runbook gap detected

10. Risk & Governance Considerations:

- Risk Mitigation
- Privilege Creep
- Access Reviews
- Insider Threat
- Timely De-Provisioning
- Unauthorized Access
- Approval Validation
- Audit Failure
- Documentation & SOPs
- SLA Breach
- Timely Acknowledgment



11. Key Learnings:

- IAM is a security-critical function
- Process discipline is essential
- Documentation protects both security and analysts
- L1 role is execution-critical in banking IAM

12. Project Outcome:

This project demonstrates readiness for an IAM / Cybersecurity L1 role by showcasing:

- Security-first mindset
- Strong SOP adherence
- Audit & compliance awareness
- Clear escalation judgment



Appendix A:

1.Joiner Process - User Provisioning:

Objective: To onboard new users securely by creating user account and assigning only the required access needed for their job role, while ensuring compliance with least privilege, audit and security policies.

Process Followed:

- Received and validated the Joiner request with documented approvals
- Created a new user account in Microsoft Entra ID (Azure AD)
- Assigned the user to appropriate security groups based on role
- Assigned RBAC roles (if required) following least privilege
- Verified user creation and access assignments
- Confirmed activity via Azure Audit Logs

Security Controls Enforced:

- Least Privilege Principle
- Role-Based Access Control (RBAC)
- Segregation of Duties (SoD)
- Approval-based provisioning
- Audit logging and traceability



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Create new user ...

Create a new internal user in your organization

Basics Properties Assignments Review + create

Create a new user in your organization. This user will have a user name like alice@contoso.com. [Learn more](#) 

Identity

User principal name * @ 

Domain not listed? [Learn more](#) 

Mail nickname *

☒ Derive from user principal name

Display name *

Password *  

☒ Auto-generate password

Account enabled ⓘ ☒

[Review + create](#)

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Create new user ...

Create a new internal user in your organization

Basics

Properties

Assignments

Review + create

Make up to 20 group or role assignments. You can only add a user to a maximum of 1 administrative unit.

+ Add administrative unit

+ Add group

+ Add role

Type	Name
Group	Finance-App-Read
Role	Reports Reader



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Create new user

Create a new internal user in your organization

Basics

User principal name	Jake_Peralta@buddhamahidhar22gmail.onmicrosoft.com
Display name	Peralata
Mail nickname	Jake_Peralta
Password	
Account enabled	Yes

Properties

User type	Member
-----------	--------

Assignments

Administrative units	
Groups	Finance-App-Read
Roles	Reports Reader

Create

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Appendix B:

2. Mover Process:

Objective:

The mover process ensures that when a user changes roles, teams, or departments within the organization, their access is updated accurately to reflect their new responsibilities. This process is critical to prevent privilege creep, maintain least privilege, and ensure audit and compliance requirements are met.

Trigger:

The mover process is initiated when:

- A user changes job role or department
- A user transitions to a new project or team
- A user's responsibilities change requiring different access
- The request is typically raised through a service request with documented approval from the user's manager and application/data owners.



Process Flow:

- Request Validation
- Validate the service request and confirm documented approvals.
- Verify the user's current role and new role details.
- Ensure approvals align with organizational IAM policies.
- Review Existing Access
- Identify current group memberships and assigned roles.
- Assess whether existing access is still required for the new role.
- Mark any excess or conflicting access for removal.

Access Revocation (Old Role):

- Remove the user from groups associated with the previous role (e.g., Finance-App-Read).
- Revoke any access that is no longer required.
- This step is performed before granting new access to avoid privilege overlap.



Access Provisioning (New Role):

- Add the user to appropriate security groups aligned with the new role (e.g., Application-Management).
- Grant access strictly based on RBAC and least privilege.
- If the new role requires elevated responsibilities, assign directory roles only with proper approval.
- Verification & Confirmation
- Confirm successful removal and addition of group memberships.
- Validate updated access aligns with the user's new responsibilities.
- Ensure no unauthorized or excessive permissions remain.

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Peralata | Groups

User

Search

+ Add memberships X Remove memberships Refresh Columns Got feedback?

Overview

Audit logs

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Applications

Licenses

Devices

Azure role assignments

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Search groups

Add filters

Name	Object Id	Group Type	Membership Type	Email	Source
☐ Finance-App-Read	7569d83b-50b1-4452-bfa3-fe3...	Security	Assigned		Cloud



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Peralata | Groups

Search

+ Add memberships X Remove memberships Refresh Columns Got feedback?

Remove member Peralata from group Finance-App-Read?

OK Cancel

	Email	Source
☒ Finance-App-Read 7569d83b-50b1-4452-bfa3-fe3... Security Assigned		Cloud

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New support request

Add or remove favorites by pressing Ctrl+Shift+F

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1

Successfully removed group membership

Membership has been removed from Finance-App-Read. Recent changes may take some time to appear.



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Peralata | Groups

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Try changing or adding filters if you don't see what you're looking for.

Search

2 results found

	Name	Email
☒	Application Management	NorthwestSales@buddhamahidhar22gmail.onmicrosoft.com
☐	Finance-App-Read	

Selected groups (1)

Reset

Application Management
NorthwestSales@buddhamahidhar22gmail...

Select

buddhamahidhar22@g... CYBERMAHI (BUDDHAMAHIDHA...)

2

Successfully added group membership

Membership has been added to Application Management. Recent changes may take some time to appear.



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Peralata | Groups

User

Search

+ Add memberships Remove memberships Refresh Columns Got feedback?

Search groups

	Name	Object Id	Group Type	Membership Type	Email	Source
☐	AM Application Management	52805515-835d-43f8-aa3f-f5ab...	Microsoft 365	Assigned	NorthwestSales@buddhamahid...	Cloud

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User has been successfully moved to the application management group from Finance-app-read.

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Peralata | Assigned roles

User

Search

+ Add assignments Remove assignments Refresh Got feedback?

Remove selected assignment(s)?

Yes No

Role	Description	Resource Name	Resource Type	Assignment Path	Type
☒ Reports Reader	Can read sign-in and audit reports.	Directory	Organization	Direct	Built-in

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Peralata | Assigned roles

User

Search

+ Add assignments

Remove assignments

Refresh

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Administrative roles

Administrative roles can be used to grant access to Microsoft Entra ID

Search by name or description

Add filters

No directory roles assigned.

Directory roles

To assign custom roles to a user, your organization needs Microsoft Entra ID Premium P1 or P2.

Choose admin roles that you want to assign to this user. [Learn more](#)

application

Add filters

Role	Description
☒ Application Administrator	Can create and manage all aspects of app registrations and enterprise apps.
☐ Application Developer	Can create application registrations independent of the 'Users can register applications' setting.
☐ Attribute Provisioning Administrator	Read and edit the provisioning configuration of all active custom security attributes for an application.
☐ Attribute Provisioning Reader	Read the provisioning configuration of all active custom security attributes for an application.
☐ Cloud Application Administrator	Can create and manage all aspects of app registrations and enterprise apps except App Proxy.
☐ Directory Readers	Can read basic directory information. Commonly used to grant directory read access to applications and guests.
☐ Directory Writers	Can read and write basic directory information. For granting access to applications, not intended for users.
☐ Network Administrator	Can manage network locations and review enterprise network design insights for Microsoft 365 Software as a Service applications.

Add

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+ Add assignments

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Administrative roles

Administrative roles can be used to grant access to Microsoft Entra ID and other Microsoft services. [Learn more](#)

Search by name or description

Add filters

Role	Description	Resource Name	Resource Type	Assignment Path	Type
☐ Application Administrator	Can create and manage all aspects of app registrations and enterprise apps.	Directory	Organization	Direct	Built-in

Successfully added assignment

Successfully added assignment 'Application Administrator'.

Add or remove favorites by pressing Ctrl+Shift+F

**Note on Role Assignment:**

In this scenario, a directory role was assigned only because the user transitioned into an application support function requiring administrative capabilities. Directory roles were assigned strictly based on documented approval and aligned with least privilege principles. In production environments, such roles are typically managed via Privileged Identity Management (PIM) with time-bound access.



Appendix C:

3. Leaver Process - De-Provisioning:

Objective: To ensure timely removal of access when a user exits the organization, preventing unauthorized access and insider risk.

Process Followed:

- Verified termination trigger (HR / Leaver event)
- Identified user account in Microsoft Entra ID
- Removed group memberships and role assignments
- Deleted user account as per organization policy
- Verified action via Azure Audit Logs

Security Controls Enforced:

- Least Privilege
- Insider Threat Prevention
- Audit & Compliance Traceability



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Home >

Adele Vance
User

Search

Edit properties Delete Refresh Reset password Revoke sessions Manage view Got feedback?

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Basic info

AV

Adele Vance
AdeleV@buddhamahidhar22gmail.onmicrosoft.com
Member

User principal name

AdeleV@buddhamahidhar22gmail.onmicrosoft.com

Group memberships

1

Object ID

f91c6321-657b-43f4-8ab5-2f22faac94c2

Applications

0

Created date time

Dec 21, 2025, 6:52 PM

Assigned roles

1

User type

Member

Assigned licenses

0

Identities

buddhamahidhar22gmail.onmicrosoft.com

My Feed

Account status

Disabled

Edit

B2B invitation

Convert to external user

Add or remove favorites by pressing Ctrl+Shift+F

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Adele Vance
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Basic info

AV

Adele Vance
AdeleV@buddhamahidhar22gmail.onmicrosoft.com
Member

User principal name

AdeleV@buddhamahidhar22gmail.onmicrosoft.com

Object ID

f91c6321-657b-43f4-8ab5-2f22faac94c2

Created date time

Dec 21, 2025, 6:52 PM

User type

Member

Identities

buddhamahidhar22gmail.onmicrosoft.com

My Feed

Account status

Disabled

B2B invitation

Delete Adele Vance

This user is an owner of the assets listed below. We recommend adding another owner before the user is deleted. Otherwise it might be difficult to determine which assets need more owners assigned to meet your organization's policies.

Deleting a user

Deleting will remove all the user's permissions and access. The user can be reinstated up to 30 days after deletion.

Name	Type
Adele Vance	Member

Associated assets

Please reassign ownership for these assets to avoid orphaned assets.

1 Role(s)

0 Group(s)

Delete

Cancel

Add or remove favorites by pressing Ctrl+Shift+F



Notifications



[More events in the activity log →](#)

[Dismiss all](#)



User successfully deleted



Adele Vance has been deleted.

a few seconds ago

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Users | Audit logs

CyberMahi

All users

Audit logs

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Deleted users

Password reset

User settings

Bulk operation results

Bulk operation results (Preview)

New support request

Download Refresh Columns Got feedback?

This view will soon be replaced with a view that includes custom security attribute logs, infinite scrolling, and column reordering. Try out our new audits preview. →

Date : Last 7 days

Show dates as : Local

Service : All

Category : UserManagement

Activity : All

Add filters

Date	Service	Category	Activity	Status	Status reason	Target(s)	Initiated by (actor)
1/22/2026, 7:14:59 PM	Core Directory	UserManagement	Delete user	Success		f91c6321657b43f48ab5...	buddhamahidhar22_g...
1/22/2026, 7:09:38 PM	Core Directory	UserManagement	Add user	Success		Jake_Peralta@buddha...	buddhamahidhar22_g...
1/22/2026, 7:09:38 PM	Core Directory	UserManagement	Update PasswordProfile	Success		Jake_Peralta@buddha...	buddhamahidhar22_g...
1/22/2026, 7:00:41 PM	Core Directory	UserManagement	Update user	Success		AdeleV@buddhamahid...	buddhamahidhar22_g...
1/22/2026, 7:00:41 PM	Core Directory	UserManagement	Disable account	Success		AdeleV@buddhamahid...	buddhamahidhar22_g...